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QUIZ B-A (EE-1005)
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SECTION: G

1. Draw Truth table for JK-FF and generate Excitation table for JK-FF and perform all required steps? [2]

Q_t	J	K	Q_{t+1}	$Q_t \rightarrow Q_{t+1}$	J	K
0	0	0	0	0 → 0	0	X
0	0	1	0	0 → 1	1	X
0	1	0	1	1 → 0	X	1
0	1	1	1	1 → 1	X	0
1	0	0	1			
1	0	1	0			
1	1	0	0			
1	1	1	1			

$Q_{t+1} = JQ_t + \bar{K}Q_t$

excitation table

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2. Perform Analysis of the circuit A given

$$Z = X'Q_1Q_0$$

J ₁	K ₁	J ₀	K ₀
$X' + Q_0$	XQ_1	$X + Q_1$	X

[10]

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$$2^{n+m}$$

$$2^{1+2}$$

$$= 2^3$$

$$= 8$$

X	Old State		J ₁	K ₁	J ₀	K ₀	New State		Z
	Q ₁	Q ₀					Q ₁	Q ₀	
0	0	0	-	0	0	0	1	0	0
0	0	1	-	0	0	0	1	1	0
0	1	0	0	0	1	0	0	0	0
0	1	1	0	0	1	0	0	1	0
1	0	0	0	1	0	1	0	0	0
1	0	1	0	1	0	1	0	1	0
1	1	0	1	1	0	1	1	0	0
1	1	1	1	1	0	1	1	1	0

State diagram on page 18

State Diagram

